

ECE330 – Fall 2024
Exam 1
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90 min

Wednesday, October 2, 2024

NAME _____

NetID _____

Instructions:

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your NetID.
- This exam is closed book and closed notes.
- To get credit, you must SHOW ALL your work and provide explanations where needed.
- You may write on both sides of the exam paper. If there is anything important on the reverse side, please indicate this on the front side.
- All answers must INCLUDE UNITS whenever appropriate.
- Graphing and/or 4-function calculators are permitted, but no cellphones/laptops.
- This exam booklet contains 9 pages (including this one) with 4 problems.

Problem 1 (25 Points)

A single-phase source is serving three loads connected in parallel:

- Load 1 draws 100 A at a lagging PF of 0.8.
- Load 2 draws 8 kW at a lagging PF of 0.7.
- Load 3 is an impedance of $(5+j2) \Omega$.

The voltage across the loads is $v(t)_{load} = 283 \sin(377t)$ V.

The impedance of **each of the two wires** that serves this combination of loads is $(0.1+j0.1) \Omega$.

a) Find the total average and reactive powers provided by the source.

P = _____

Q = _____

b) Find the source voltage and current as time domain quantities.

$$i(t)_s = \underline{\hspace{15cm}}$$

$$v(t)_s = \underline{\hspace{15cm}}$$

c) What is the combined power factor of the three loads if we measure it at the load side? State whether it is leading or lagging.

$$\text{Combined PF} = \underline{\hspace{15cm}}$$

Circle one: Leading / Lagging

Problem 2 (25 Points)

A 3-phase, 208 V (line-line), Wye-connected 60 Hz source supplies 5A (magnitude) of line current to a 1500 W, 3-phase wye-connected load at a lagging power factor.

- a) When a new wye-connected purely reactive load is connected in parallel with the 1500 W 3-phase load, the line current (magnitude) increases to 7A. What is the impedance per phase of the new wye-connected load?

Hint: You may assume that the real power does not change when the new wye-connected load is added.

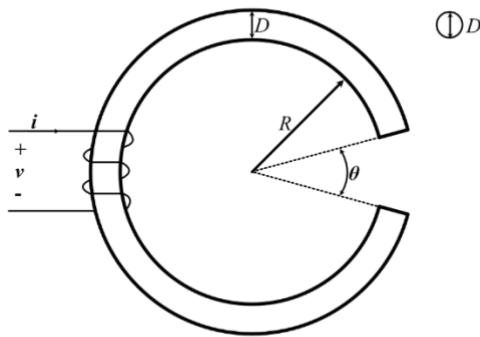
$$\bar{Z}_Y = \underline{\hspace{15cm}}$$

b) Now a delta-connected capacitor bank is added in parallel with the two loads, reducing the total line current from 7A to 4.75 A (magnitude). What is the size of the capacitors added per phase in Farads (capacitance) in the delta-connected capacitor bank?

$C =$ _____

Problem 3 (25 Points)

A toroidal iron core with permeability $\mu = 1500\mu_0$ and inner radius $R=10\text{cm}$, shown below, has a coil with 100 turns wrapped around it. The cross section of the toroid is a circle, as shown, and has a diameter $D=2\text{ cm}$. There is a cut in the toroid that spans an angle $\theta=10^\circ$. Fringing effects can be neglected.



a) What are the reluctances for the iron core and the air gap?

Hint: the arc length of an arc lying on a circle with radius r spanning θ radians is $l = r\theta$.

$$R_{core} = \underline{\hspace{15cm}}$$

$$R_{gap} = \underline{\hspace{15cm}}$$

b) What is the total magnetic flux in terms of unknown current i ?

$$\phi = \underline{\hspace{15cm}}$$

c) What is the inductance of this system?

$$L = \underline{\hspace{15cm}}$$

Problem 4 (25 Points)

Note: Parts **a)** and **b)** are unrelated.

- a)** An ideal 240V/120V, 60 Hz single-phase transformer is being used to supply power to a load drawing 1kW at 0.8 PF lagging. Note that 240V corresponds to the primary side, and 120V corresponds to the secondary side.

- i. Find the complex power being supplied to the load.

$$\bar{S}_L = \underline{\hspace{15cm}}$$

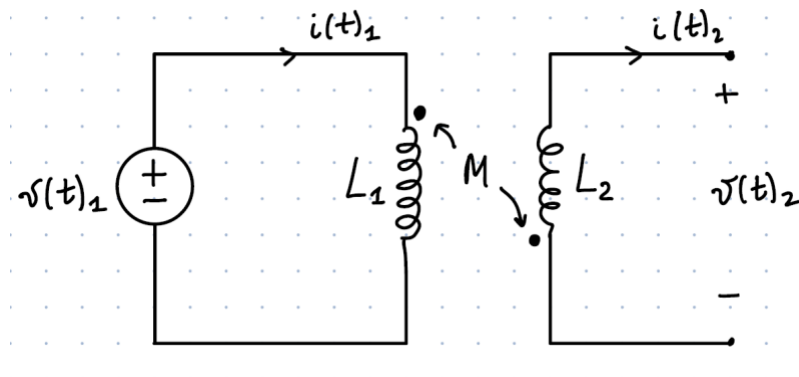
- ii. Find the load impedance.

$$\bar{Z}_L = \underline{\hspace{15cm}}$$

- iii. Find the load impedance as seen from the primary side.

$$\bar{Z}_p = \underline{\hspace{15cm}}$$

b) Consider two coils magnetically coupled together as shown on the diagram below.



Self-inductances of the coils are given as $L_1 = 2 \text{ H}$ and $L_2 = 250 \text{ H}$. Mutual inductance between two coils is given as $M = 5 \text{ H}$. Coil 2 is open-circuited. What is the voltage $v(t)_2$ when a sinusoidal voltage $v(t)_1 = 1\sqrt{2} \cos(377t) \text{ V}$ is applied to coil 1?

$$v(t)_2 = \underline{\hspace{15cm}}$$